RN information in the original variables

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Contents

Mathematical foundation and source contract **1**

Validity by finite-box projection . . . 2
Numerical transport . . . 2
Matched-root value and its limits . . . 2
Once-root integration and scope . . . 3
Original-space structure and scalar floors . . . 3
Saved-point witnesses and provenance . . . 4

Matched LP and complete-method evidence **4**

Information retained by one original-variable row . . . 4
Structural content and scalar floors . . . 5
Complete same-PE results . . . 5
Implementation, qualification and limits . . . 8
Cost, adoption and remaining scope . . . 8

References **9**

Mathematical foundation and source contract

The experimental question is whether one cheap root RN source can retain useful normalized Gini information without retaining the RN columns in the formal GI-TBC models. Stable R137, its own H, M-B/VD-P, pair-only support, AM=.08, depth 8/minimum width 1e-4, AP off and R134 proof policy remain the carrier. The research flag defaults off.

Validity by finite-box projection

Let the matched continuous source be

$$\min\{c^T x + c_0 : x \in P,\ Ex + Fz \geq d,\ \ell \leq z \leq u\}.$$

Only RN rows belong to this block; old routing rows, the objective cutoff and the old objective remain in P. Equalities have unrestricted multipliers. All new RN columns have objective coefficient zero. Production-redundant finite caps are retained: $t \in [L_t, U_t]$, $w_{iy} \in [0, U_t]$, normalized inventory $\widehat{Y}_i \in [0, M_i U_t]$ (outward rounded) and normalized pair epigraph $\widehat{h}_{ij} \in [0, 1]$. The old diagnostic's infinite upper bounds are never interpreted as finite.

For any legal multiplier vector β, set $a = E^T\beta$, $r = F^T\beta$ and $b = d^T\beta$. Then

$$a^T x \geq b - \sigma_{[\ell,u]}(r), \qquad \sigma_{[\ell,u]}(r) = \sum_j \max(r_j \ell_j, r_j u_j).$$

This validity argument needs no dual optimality and no exactly zero floating residual. It applies to the actual RN row coefficients and actual finite bounds. The resulting row contains only original selectors and G.

Numerical transport

The implementation uses the equivalent $\leq$ convention. A $\geq$ RN row is negated and weighted by $\max(\Pi, 0)$; a $\leq$ row is weighted by $\max(-\Pi, 0)$; an equality is oriented according to its unrestricted Π and weighted by $|\Pi|$. Wrong-sign inequality multipliers are clipped to the legal half-axis and recorded. Thus the combination is $A_x x + A_z z \leq B$. Zero is submitted on every extension column. The signed finite-box conversion bound produces

$$\widehat{A}x \leq \widehat{B}, \quad \widehat{B} \geq B + \sum_j \max_{v \in [l_j, u_j]} (\widehat{A}_j - A_j)v.$$

For eliminated columns, this is precisely $-\min_{\ell \leq z \leq u} A_z z$. For retained columns, it compensates coefficient rounding, including deletion below 1e-12. `projectObjectiveResources` encloses every product/addition with directed binary64 intervals and rounds the final $\leq$ RHS upward. Existing R104 APIs and behavior are preserved. Independent Fraction checks interpret every binary64 row, multiplier, bound and submitted coefficient exactly. Actual Native readback checks all terms and RHS; compact LP reoptimization measures the remaining numerical loss.

Matched-root value and its limits

Under exact optimal primal/dual solutions, the RN part of the optimal dual attains the extension-box support at the optimal z. Replacing that dual block by its single projected row retains the same dual lower bound; the RN optimal x also satisfies the row. Hence the two matched root optima are equal. These are the same-domain, same-objective assumptions of standard LP duality and the earlier R104 result. This is not a new duality theorem or the complete RN projection. A narrower compatible descendant inherits validity; its optimum need not match RN. Grouping before box support can lose cancellation, since $\sigma(r_1) + \sigma(r_2) \geq \sigma(r_1 + r_2)$. No grouping grid is pursued.

R104's ACTIVE root preservation accompanied full-method regression; its ACTIVE and SHADOW each used 4349 auxiliary LPs and 3644 DPs. Those negative results remain. RN-P instead pays once for the newly established R143 RN information and removes the extension columns from the formal model. Neither the old result nor the compression theorem predicts search speed.

Once-root integration and scope

The source runs after the current run's own H and initial root are known. Its artifact, Native call, X/Pi/RC/basis, source cost and status have separate identities. The source handle is discarded; its point and bound are not installed as formal control or Start. The original model is rebuilt/embedded with the PC and optimized normally. Future artifacts accept the PC only for the same input and contained G/cutoff/support domains. An absent root selector in a narrower support is exactly zero. New runs clear all source/cache state. No instance identity, size, leaf-time threshold or hardware clock chooses the representation. A zero RN dual aggregate gives an exact $0 \le 0$; its outward transported tautology $0 \le B$ with $B \ge 0$ is omitted as a mathematical redundancy. A nonpositive proved denominator bound omits the RN source uniformly. Failed/unknown sources retain their actual status.

Original-space structure and scalar floors

A scalar objective floor can also match a single root value. Structural information is distinguished here on an explicitly limited rational block. For any rank support a of the pairwise absolute-difference sum, write the normalized inventory as $v_i = c_i \sum_y y w_{iy}$. With $\sum_i v_i = 1$, $\sum_y w_{iy} = t$, and $L_t s_{iy} \le w_{iy} \le U_t s_{iy}$, any η and station multipliers λ_i give

$$d_{iy} = (a_i - \eta) c_i y - \lambda_i, \qquad \kappa_{iy} = \min(L_t d_{iy}, U_t d_{iy}), \quad \Lambda = \sum_i \lambda_i,$$

$$ng \ge \sum_i a_i v_i = \eta + \Lambda t + \sum_{i,y} d_{iy} w_{iy} \ge \eta + \min(L_t \Lambda, U_t \Lambda) + \sum_{i,y} \kappa_{iy} s_{iy}.$$

This Rank-PC family is valid independently of its strength at a chosen point. The station coefficients are the minimum of two affine functions of the inventory state. A rank vector has entries $2\,\mathrm{rank}_i - n - 1$, or a convex combination of such permutations; this is the ordinary support function of the absolute-difference sum.

Consider the continuous three-station block with $c_i = 1$, supports $\{0, 1\}$, inventory sum in [1,2], and selector mass at inventory one fixed to (1,1/2,0). This is a declared LP cross-section, not an integer routing domain. The old R143 witness has $g = 1/3$ and old g-state products $q = ((0, 1/3), (0, 1/3), (1/3, 0))$. Each station's products sum to g and obey the full unit-box product bounds. Here $H = 2$, $J = 2/3$ and $\tau = 3/2$; $H \le 3J$, TC $H \ge \tau$, TC2 $H \ge 2$ and the stock bounds

$$\max(g, \tau + 2g - 2) \le J \le \min(\tau + g - 1, 2g)$$

all hold. Endpoint pair couplings attain the same absolute differences because each pair has at least one fixed inventory marginal. With rank $a = (2, 0, -2)$, $\eta = 1$ and station multipliers $(1, -1, 0)$, the valid Rank-PC is

$$3g \ge 1 - s_{1,0} + \tfrac{1}{2} s_{2,0} - 3 s_{3,1}.$$

Its RHS at that point is $5/4$, so g must be at least $5/12$. The RN optimum in this fixed-selector cross-section is exactly $5/12$: normalization gives $v = (t, 1 - t, 0)$, while the middle zero-state product is $2t - 1 \ge 1/4$.

Thus $t \geq 5/8$ and $g \geq 2t/3 \geq 5/12$. Equality is attained at $t = 5/8$ with $w = ((0, 5/8), (1/4, 3/8), (5/8, 0))$; choosing old $q = ((0, g), (0, g), (g, 0))$ also satisfies the retained old rows at $g = 5/12$.

Append an independent $p \geq 0$ and use the same objective $g + p$ in both old and RN blocks. The RN optimum is still $5/12$ (attained with $p = 0$), so this is a valid matched scalar floor. At the old point, $p = 1/12$ gives $g + p = 5/12$: the scalar floor is satisfied while the structural PC remains violated by 1/4. This witness is confined to the declared old Gini/stock/count/endpoint-product block with that independent epigraph. It is not a strict separation witness for an entire routing P, and it proves no universal floor dominance.

Saved-point witnesses and provenance

The six goal-supplied Rank-PC point violations have been finitely reconstructed from saved R143 ranks, eta, supports, denominator bounds and old points, with zero Optimize. Violations divided by n are .023855359195111166, .005944242130579499 and .011385760529762846 for K35 BASE, stock and TC+stock; they are .06850590621823809, .022542692207833517 and .022542692207833594 for the corresponding RX20-T1 domains. The reconstructed legal station subgradients give exact nonzero counts 15/15/40 and 14/14/14. These are their own certificate identities, not the absent supplemental ZIP's exact rows with counts 19/19/44 and 14/14/14. The point violations agree with the goal's displayed decimals and are not newly optimized objective gains. R143's old raw has no complete Pi/RC/basis; this round's paid sources genuinely obtain those attributes. New source/compact LPs and compiled-row qualification are reported separately from those point certificates and from full-method costs.

Matched LP and complete-method evidence

The decision is **RESEARCH_ONLY**. One full RN-block dual box projection retains essentially all measured matched-root RN gain without persistent RN columns. The four-role same-PE core panel remains mixed: K35 has useful incumbent/gap gains with a weaker supported lower bound, RX20 proves the same optimum slightly sooner, K20 has no new row or search benefit, and RB35 has an important adverse complete endpoint. The K35 same-PE full-RN bridge confirms a useful representation result, without establishing broad stable-method superiority.

Stable R137 GI-TBC, its paper entry/profile and earlier research entries remain available. The RN-P binary flag defaults off; its separate research launcher explicitly selects this candidate. This is a finite representation-cost experiment, not a claim that normalized Gini information or root compression is universally fast.

Information retained by one original-variable row

Role	BASE objective	RN objective	Compiled PC objective	RN gain retained
K-35-U	0.200177661949	0.228090496924	0.228090496924	1
R-X20-S-T1	0.331594316287	0.389761616211	0.389761616211	1

Displayed table values are rounded. The actual compiled interval writer loses only 3.83026943496e-15 and 1.36557432029e-14 in matched root objectives. K35 removes 1170 persistent RN columns/2340 RN rows; RX20-T1 removes 354/708. Each useful compact model adds one selector/G row and no new columns. All old matrix rows, objective and penalty, cutoff, supports and G domain are retained. The two RN source LPs newly save optimal source Pi/RC/basis; the other two primary LPs reoptimize their matched compact models. R143's raw did not contain those complete dual attributes. Two additional compact optimizations qualify the actual C++ coefficients after Native transport.

The single saved actual RX20 upper AM-lookahead child has inherited-PC and full-RN objectives both 0.588107599905. Only its RN comparison is newly optimized; the compact optimization is reused from the sealed health qualifier. There are seven new standalone LP solves in total. This one no-loss child does not establish full-tree equality.

Box support and original-box coefficient compensation establish validity for actual binary64 multipliers, including residual extension coefficients and tiny-term deletion. Independent Fraction checks and actual Native readback verify the final rows. Exact same-root optimal-value preservation requires exact optimal primal/dual solutions, the same P and objective. It is the standard dual-aggregation principle already present in R104; it is neither a new theorem nor the complete RN projection. R104 ACTIVE's full regression and its 4349 auxiliary LPs / 3644 DPs per ACTIVE/SHADOW remain prior negative evidence.

Structural content and scalar floors

On the declared three-station fixed-selector continuous Gini/stock/count/product cross-section, the RN optimum for objective g+p, with independent p>=0, is exactly 5/12. The old point g=1/3, p=1/12 satisfies that matched scalar floor; the selector/G Rank-PC still excludes it by 1/4. Exact old-row feasibility, the RN lower proof and an attaining extension are retained in the local witness. This strictly distinguishes the two on that declared local block. There is no claimed exact separating point for the complete routing P, no universal floor dominance and no use of floating primal ObjVal as a paid proof floor. The six saved-point Rank-PC violations were finitely reconstructed without Optimize and agree with the goal's displayed decimals. Different legal station subgradients give their own certificate identities and sparsities; they are not represented as byte-identical to the absent supplemental ZIP.

Complete same-PE results

Role / method	U	Supported L	Relative gap	Certificate
K-20-S / OFF	0.716806546278	0.705363448597	0.0159639971768	False
K-20-S / RNP	0.716806546278	0.705346769145	0.015987266288	False
K-35-U / OFF	0.692854358631	0.558227874949	0.194307045926	False
K-35-U / RNP	0.660923637174	0.545412294185	0.174772600784	False
K-35-U / RN	0.681258498381	0.542931774252	0.203045869458	False
R-X20-S-T1 / OFF	0.645325030386	0.645325030386	3.44081810669e-16	True
R-X20-S-T1 / RNP	0.645325030386	0.645325030386	2.70104221375e-14	True
R-B35-S-T2 / OFF	0.344642625592	0.218985949603	0.364599926584	False
R-B35-S-T2 / RNP	0.362136826572	0.218895658669	0.395544328533	False

Role / method	T_produce	T_cert	T_verify	T_total
K-20-S / OFF	1172.37793		286.738826	1459.11676
K-20-S / RNP	1173.51913		61.6877434	1235.20687
K-35-U / OFF	2371.08262		3.9850276	2375.06765
K-35-U / RNP	2371.13836		5.1892433	2376.3276
K-35-U / RN	2371.02396		7.7170262	2378.74099
R-X20-S-T1 / OFF	20.6335916	20.6335916	1.1063039	21.7398955
R-X20-S-T1 / RNP	18.9138269	18.9138269	1.4015167	20.3153436
R-B35-S-T2 / OFF	2370.99042		3.8214044	2374.81183
R-B35-S-T2 / RNP	2371.02124		5.1370398	2376.15828

Times are seconds. Open T_cert entries are intentionally empty. T_produce includes each run's own H, the once-root source where applicable, row construction, formal solves and in-process checks. T_verify contains all mandatory sealed-raw postverification; T_total is their actual complete clock. Same-role pairs verify equal physical inputs/fleets, H semantics and H fleet, source/PE, Seed and full cap. Native return ObjBoundC and

the external merged supported L remain separate fields in reports/new_arms.csv; signed U-L is retained without clipping. No historical fleet, point or bound is injected.

Role / method	Presolved columns	Terminal nodes	Terminal Work	Source prep s
K-20-S / OFF	2884	450091	1907.09124	0
K-20-S / RNP	2884	448888	1901.88085	0.4785896
K-35-U / OFF	7647	17076	4724.21697	0
K-35-U / RNP	7647	14826	4843.21752	1.7341766
K-35-U / RN	8803	11386	5336.40045	0
R-X20-S-T1 / OFF	1369	2621	29.5418052	0
R-X20-S-T1 / RNP	1369	2079	26.060295	0.3132779
R-B35-S-T2 / OFF	9994	6707	5793.33201	0
R-B35-S-T2 / RNP	9994	6705	5744.80916	2.3387603

Nodes, Work, simplex, presolved size, AM decisions and trajectories describe the observed searches. Fewer nodes or equal LP roots alone do not establish complete benefit, and Work/node is not a causal measurement of column cost.

K35 RN-P improves physical U by 4.6086% and relative gap by 1.9534 percentage points versus OFF, while supported L falls 2.2958%. Its presolved columns remain 7647, root simplex iterations fall from 5188 to 3794, and the printed root-after-cuts bound rises from .42856 to .44959. Terminal Work nevertheless rises 2.5189%; this is useful mixed complete evidence, not a monotone bound or speed improvement.

RX20-T1 proves the same .6453250303858205 physical objective in both methods. RN-P producer/certificate time is lower by 1.7198 seconds and complete time by 1.4246 seconds. This small measured benefit is retained without a general timing claim.

K20 and RB35 both produce the exact tautology 0<=0, so the uniform rule emits no PC. Their once-source preparation costs are .4785896 and 2.3387603 seconds. K20 has identical U, a lower L by 1.66795e-5, and producer time higher by 1.1412 seconds. Its actual pure verification times are 286.7388 seconds OFF and 61.6877 seconds RN-P, giving RN-P a 223.9099-second lower observed T_total. These clocks remain unchanged; no structural search acceleration is inferred on a no-PC role. RB35 RN-P has U worse by 5.0760%, L lower by 9.02909e-5 and gap worse by 3.0944 percentage points. The verified trajectories largely coincide before OFF's late incumbent improvement; this important adverse candidate endpoint is preserved, with no claimed structural cause or proof of global RN redundancy.

The actual same-PE K35 full-RN bridge has U=.6812584983808189, L=.5429317742515352 and gap=.20304586945785152. RN-P improves its U by 2.9849%, L by .4569% and gap by 2.8273 percentage points. RN-P removes 1170 formal RN columns; presolved columns fall from 8803 to 7647, root simplex iterations from 22953 to 3794, and terminal Work from 5336.4005 to 4843.2175, despite more nodes (14826 versus 11386). RN-P T_produce is .1144 seconds higher and T_total 2.4134 seconds lower, mainly through verification. These are complete capped-window endpoint and representation observations, not a causal isolation of column count or an earlier proof. Root-after-cuts bounds .44900 and .44959 show the RN information is not immediately erased by Native cuts.

The figure uses postqualified committed physical witnesses and their effective availability on each producer wall clock. It does not relabel Native log time as T_produce or T_total. Full supported L, signed gaps and complete verification costs remain in the tables and full-precision reports.

RB35-T2 emits no PC: its formal model and own-H identity match OFF. The adverse endpoint remains part of the complete candidate's evidence. This verified trajectory records the late OFF improvement; it establishes no structural cause or general redundancy of RN.

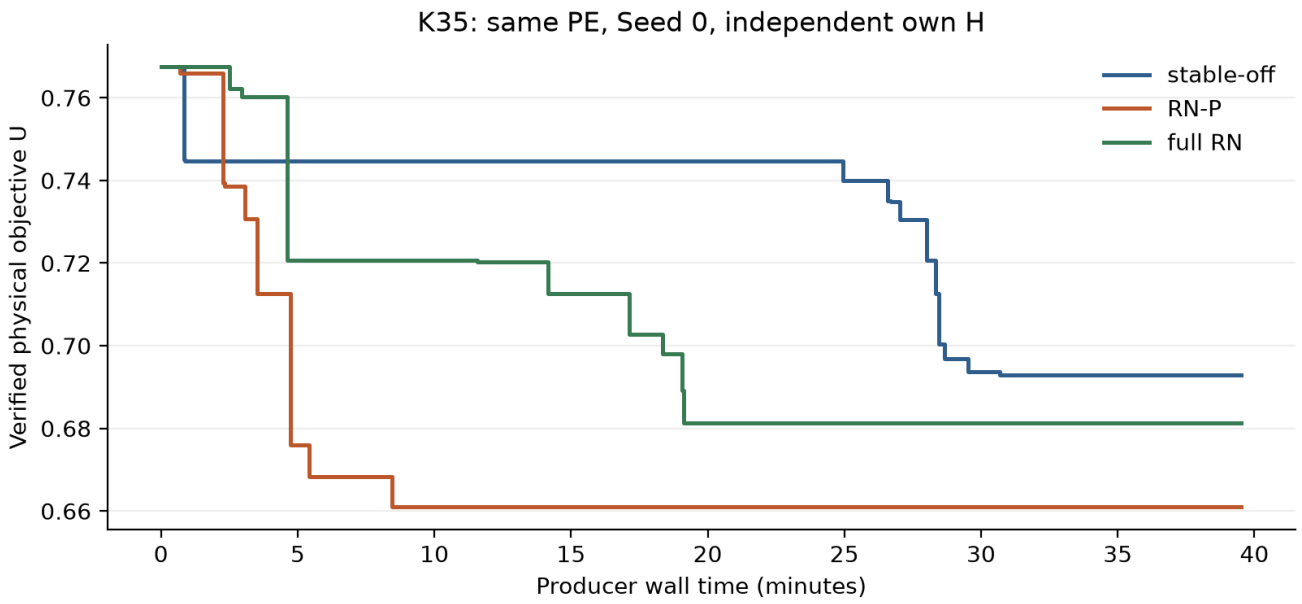


Figure 1: K35 physical incumbent trajectories

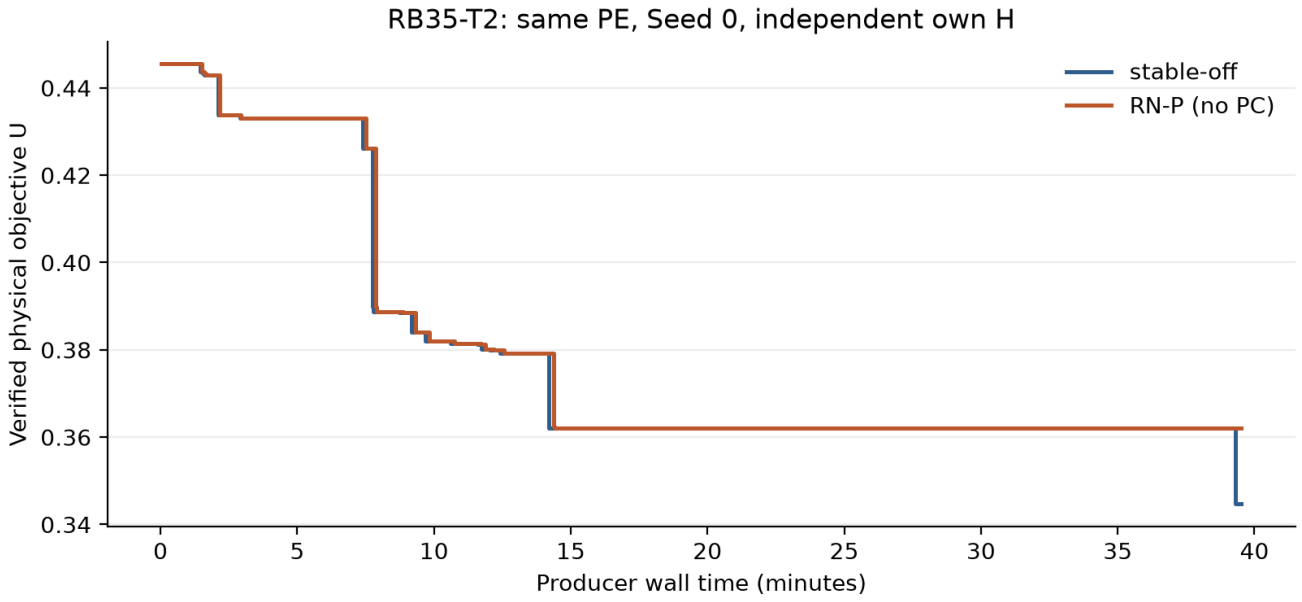


Figure 2: RB35-T2 physical incumbent trajectories

Implementation, qualification and limits

RN-P obtains one production-capped RN continuous source after the run's own H and initial root are known. Actual mixed row directions and scaling are checked. The source's types are restored and its handle explicitly discarded; formal models use distinct canonical hashes, original integer columns and unchanged Start mapping. Only compatible narrower input/G/cutoff/support domains inherit the PC. All source calls remain in Native quality, lifecycle and cost audits; they are excluded from formal controller-bound reconstruction. A zero RN dual block is a uniformly omitted tautology. Nonpositive proved denominator bounds uniformly omit the source. Every complete run clears its source/cache state; no instance/size/time-specific rule is introduced.

The initial K20 qualifier failed on a zero aggregate before durable dual serialization. A second K20 producer exposed the disposable source's missing integer-restoration lifecycle count; it remains unqualified. V5 restores the source types before disposal. RX20's active-PC health producer then passed; reader-only formal/source ledger separation was recovered from sealed raw, without repeating Optimize. All failed producers/readers and costs remain. Delayed health-verification latency is retained but is not a performance cell. The complete-run/qualification ledgers contain 55 observed Native Optimize calls, all returned; 7 belong to the two unqualified attempts. These counts include once-root sources but exclude the seven separately reported standalone LPs. Reader-only recoveries perform zero Optimize and retain their actual outer costs. Compiled mixed-sense/free-equality/sign-repair/signed-box/deletion/reset fixtures, actual writer/Native rows, physical complete Start, off matrix and scope rejection checks are preserved. Independent pre-full and final reviews have separate evidence and identities.

Measured production source is `0b5858544a671722142c8450db9b88c09c9b1c72`; PE SHA256 is `180126ae1ecf7db892c3fbb92faa73b517d19029e9824f62cd922e0d9e277d7b`. The measured link uses verified inherited object and archive lineage and newly compiled affected units. CMake explicitly links the same projection arithmetic library for a clean build. The process and commercial runtime binaries/licenses are not published.

Cost, adoption and remaining scope

New experimental outer cost is **14727.425396300 seconds (4.090951499 hours) of 28800 seconds**. Failed qualification/readers, once-root sources and all mandatory postverification are included. Nested H/LP/MIP/Work observations are not charged a second time. Build time (151.550907 seconds), exact offline arithmetic, paper/PDF work and public I/O are separate. Pure setup/debug waiting is not invented as extra Native compute; actual delayed qualification T_total remains visible. All entered formal positions are preserved with their full windows.

The pilot's single mixed K35 endpoint did not stop expansion: all four core roles and the one meaningful K35 RN bridge received their full windows and complete verification. The overall finite panel, including an important RB35 adverse endpoint and no added root information on two roles, does not justify replacing stable R137. Retain the useful original-variable RN projection as a separate default-off research implementation and close this once-root representation experiment. Do not infer that RN or projection can never be useful elsewhere.

No paid-floor control is entered because the present adoption decision and the claimed same-PE representation comparison do not depend on a full-routing structure-versus-scalar performance claim. The exact same-objective fixed-selector local witness establishes the limited structural distinction; no strict full-routing separation, universal floor dominance or production paid-floor benefit is claimed. K15, RX35-T2 and Seed1 remain explicitly unentered. If stable adoption is reconsidered, a paired full Seed1 check of the decisive RB35 adverse role is a priority; a paid-floor active K35 control is useful if that specific structural-performance attribution is sought. Such measurements could change confidence or future direction, but a control grid is not needed to accept the current research-only scope. Unused budget is not allocated merely to use it.

The continuation/delivery record identifies actual Draft PR/base/head, fixed science, finite carrier/member recovery and post-PR current-round cleanup. Local scientific acceptance and public byte readability are separate. Retained new research assets stay under ExactEBRP-work; the original user tree and old rounds are protected. No merge, force-push, historical deletion, email or submission is made.

The linear-fractional and Gini antecedents remain standard (Charnes and Cooper 1962; Yue et al. 2013; Chen and Hooker 2023).

References

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